Dhawal Shah

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Professional Summary

Senior ML Engineer (**5+ years**) and **AI researcher**, shipping production **agentic AI/LLM** systems while co-authoring peer-reviewed **inference-optimization** research (**AAAI'27**). Builds systems that cut cost and hold up under scrutiny.

Skills

Agentic AI & Orchestration: LangGraph, LangChain, RAG, Tool-Calling, Prompt Engineering, LLM Orchestration, Vector DB (Qdrant)
Modeling & Inference: PyTorch, TensorFlow, RLHF (PPO, DPO, GRPO), Hugging Face, NLP, Computer Vision, OpenCV, Spacy, ClearML, Distributed HPC
Backend/Software Engineering: Python, FastAPI, REST APIs, PostgreSQL, MongoDB, MySQL, Alembic
Data Engineering: Spark, Hadoop, PySpark, Scala, Pandas, Databricks
Cloud/DevOps: AWS (S3, SageMaker, Comprehend, Lambda, Textract, EMR, EKS, EC2, ALB, RDS, VPC), Docker, Kubernetes, Terraform, Redis, GitLab/GitHub CI/CD

Experience

Hermamind Inc., Iowa
Founding AI Engineer **Jan 2026 – Aug 2026**

- Eliminated **LLM-invented prices** (invalid execution outcomes cut from 9 to 0 of 12 traces in an adversarial test suite) by architecting a deterministic-pricing **LangGraph** multi-agent workflow (RAG-style tool-calling, customer/product lookup, anchor pricing, quote generation) with typed **Pydantic** state
- Built a provider-agnostic **LLM** routing layer (**OpenAI, Anthropic, Ollama, Qwen**) evaluating per-step **latency, cost, and quality**, cutting inference cost **93%** while holding task-success at **100%** across benchmark tasks
- Redesigned **PostgreSQL** schema and shipped **Alembic** migrations across a multi-tenant architecture (central auth + per-tenant DBs) secured with **JWT/RBAC**, implementing **FastAPI** as the sole database interface with 0 cross-tenant breaches across isolation tests
- Instrumented the platform with **OpenTelemetry** (traces, metrics, logs) and enforced a **GitHub Actions CI/CD** pipeline with a 200+ test **pytest/Playwright** suite as PR merge gates
- Authored **Terraform** Infrastructure as Code for **AWS** (VPC, EC2, RDS, ALB), releasing staging and production environments with a repeatable, API-first cloud deployment pipeline

Avalara Technologies Pvt. Ltd.
Senior Software Engineer - Machine Learning **Feb 2020 – Aug 2024**

- Cut inference costs **90%** versus AWS Comprehend by building and deploying custom **ML models** (**Random Forest, Donut, LayoutXLM, LiLT, Spacy**) for document processing
- Built a model abstraction layer training multiple candidate models per task and auto-serving the most accurate/efficient one, benchmarked against **AWS Comprehend** on a 500-document evaluation set
- Automated **30% of 100,000+ manual document processing** across 4 teams, cutting per-document time **10x**
- Added **OAuth/SSO** authentication to an internal dashboard application for use by other teams, and tracked sprint work in **Jira**
- Built an async training-job service with **FastAPI/Redis**; mentored **8** new hires over **50+** hours

Projects

RLHF Alignment Method Comparison
COMS 6730 Course Project - Iowa State University **Aug 2025 – Dec 2025**

- Implemented **GRPO** using **Hugging Face TRL's GRPOTrainer** for a controlled comparison of **PPO**, **DPO**, and **GRPO** fine-tuning on **LLaMA-3.2-8B**, achieving a **72.8%** win rate at under 60% of PPO's compute cost

VarRate: Variable-Rate KV Cache Compression
Co-Author, AAAI'27 Submission - Iowa State University **May 2026 – Jul 2026**

- Owned the full **ablation and statistical validation suite** (14 studies) for a training-free **KV cache compression** method on long-context **LLMs** (**Llama-3.1-8B, Qwen2.5-7B**), using paired **bootstrap significance testing** with **Holm-Bonferroni correction** across 12+ model/task/regime cells to validate claims for peer review
- Ran controlled **A100** latency benchmarking, quantifying a **+42.9%** prefill overhead statistically indistinguishable from zero

Mac Pruner
Graduate Research Assistant - Department of Computer Science, Iowa State University **Dec 2025 – May 2026**

- Integrated MaskLLM gradient-free **N:M structured pruning** into a **LangGraph** multi-agent **Python** framework across **Vision Transformer** and LLaMA architectures, achieving **55%** MAC reduction with **72.31%** top-1 accuracy on DeiT-Tiny

Education

Iowa State University **Aug 2026**
Master of Science in Computer Science - Artificial Intelligence **3.9/4 GPA**

Institute of Engineering and Technology, DAVV, Indore **Apr 2020**
Bachelor of Engineering in Computer Engineering **8.84/10 CGPA**